

Proofreading notes for SEVEM_Stokes

October 4, 2026

Scope: proof correctness, missing references, and misprints. Keep the method, section order, and numerical data. Mark substantive manuscript changes with `\AI{...}`; leave routine wording and typo corrections unmarked. The proof corrections and presentation edits below have been applied to `SEVEM_Stokes.tex` and `numerics.tex`. Author checks remain listed separately. The original sources and PDF are saved in `proofreading/original/`.

Proof corrections

1. **Mesh assumptions (Section 1.1).** Star-shapedness alone does not make the face cones uniformly regular, and the center of the largest inscribed ball need not be a valid star center. Choose the center of a ball with respect to which the element is star-shaped. State the face and cone regularity, bounded face count, and diameter comparability used by the trace and scaling estimates. Use h_K locally and $h = \max_K h_K$ globally.
2. **Velocity degrees of freedom (Section 1.2).** Replace the set difference $\mathbb{P}_k(K; \mathbb{R}^d) \setminus \mathbf{G}_k(K)$ with the $L^2(K)$ -orthogonal complement $\mathbf{G}_k(K)^\perp$ within the polynomial vector space. The set difference is not a linear space and does not specify an independent set of moments. State $k \geq 1$ and the convention that negative-degree polynomial spaces are zero. Cite the unisolvence result.
3. **Stress space (Section 2.1).** The two original definitions of $\Sigma_k(T)$ differ. Extending $\mathbb{P}_k(F)$ constantly in the normal direction and adding $\mathbb{P}_{k-1}(T)$ does not give all of $\mathbb{P}_k(T)$ in the normal-normal block. For $k = 1$, it gives no normal-coordinate linear term, whereas the displayed block decomposition does. Use the block decomposition, which agrees with the stated normal bubble $\lambda_{\mathbf{x}_K} \mathbb{P}_{k-1}(T) \mathbf{n} \otimes \mathbf{n}$. This choice must be checked against the implementation.
4. **Stress unisolvence (Lemma 2.2 in the original source).** Replace the bubble set difference by its orthogonal complement to the selected tensor-polynomial subspace. Add the missing proof: zero face moments put the tensor in the bubble space; zero complementary moments put it in the selected subspace; global polynomial moments, tested with the dual tensor basis, then force every polynomial coefficient to vanish. Count dimensions to establish existence as well as uniqueness.
5. **Uniform stability (Assumption 2.1).** Spanning $\mathbb{S}$ on each element proves algebraic independence but does not give a mesh-independent constant. Require a uniformly conditioned choice of normalized face-tangential tensors. Otherwise the scaling constant in the stress inf-sup argument may diverge as the mesh changes. Prescribe tensor face moments separately on each element side, since the stress is discontinuous.
6. **Norm and Korn arguments (Section 2.2).** The original local equivalence with the full gradient is false: choose a nonzero rigid rotation and matching boundary values. The strain and projected jump vanish, but the gradient does not. Replace it by a global broken Korn inequality, including boundary faces. In the norm proof, use both zero normal and zero tangential boundary values; the statement that boundary normals span $\mathbb{R}^d$ alone does not justify eliminating an affine rigid motion.
7. **Full versus projected jump (Sections 2.2 and 3.2).** The norm controls $\Pi_{\partial K} \mathcal{J}_{\partial K}$, while the residual contains $\mathcal{J}_{\partial K}$. Add

$$h_K^{-1/2} \|\mathcal{J}_{\partial K}(\mathbf{v}_h)\|_{\partial K} \lesssim \|\mathbf{v}_h\|_{1,h,K}.$$

To prove it, subtract a local rigid motion from $\mathbf{w} = \Pi_K^o \mathbf{v}_h^{\text{div}}$, use Korn's inequality modulo rigid motions and the trace inequality, and use the fact that the tangential face space contains the tangential traces of rigid motions.

8. **Pressure stability and interpolation (Section 3).** Add proofs of the pressure continuity and right-inverse lemmas. Recall the commuting identity

$$\operatorname{div} I_K^{\text{div}} \mathbf{v} = Q_h(\operatorname{div} \mathbf{v})|_K.$$

Construct the pressure lifting by interpolating an H_0^1 divergence lifting, and bound its discrete norm. Since $b(\mathbf{v}_h, q_h) = -(\operatorname{div} \mathbf{v}_h^{\text{div}}, q_h)$, use the negative lifting when a positive inf-sup numerator is needed. Add absolute values to continuity bounds and exclude zero denominators in suprema.

9. **Error equation and approximation (Section 3.2).** Restore sums over K : cancellation of the exact traction on interior faces is a global statement. The strain of $\Pi_K^o I_K^{\text{div}} \mathbf{u}$ already belongs to $\mathbb{P}_{k-1}(K; \mathbb{S})$, so its stress projection is redundant. Also,

$$(\operatorname{div} \Pi_K^o \mathbf{v}_h^{\text{div}}, p - Q_h p)_K = 0.$$

The original estimate drops $\varepsilon(\mathbf{u} - \Pi_K^o \mathbf{u})$ without justification. Retain the polynomial approximation error and bound $\mathbf{u} - \Pi_K^o I_K^{\text{div}} \mathbf{u}$ in the volume and on faces. Estimate only the virtual interpolant's normal trace, which is a face projection; no full H^1 trace of a virtual field is needed.

10. **Meaning of the final velocity error (Section 3.3).** The norm is initially defined only on discrete pairs. Identify a smooth velocity with $\{\mathbf{u}, \Pi_{\partial K}^t \mathbf{u}\}$ before writing $\|\mathbf{u} - \mathbf{u}_h\|_{1,h}$. Extend the volume L^2 projector to all L^2 vector fields. Keep the stated $O(h^k)$ estimate; do not add a claim of pressure robustness or an L^2 velocity theorem.

References to add at the point of use

1. Beirão da Veiga, Brezzi, Marini, and Russo, *$H(\operatorname{div})$ and $H(\operatorname{curl})$ -conforming virtual element methods*, Numerische Mathematik 133 (2016), 303–332. [doi:10.1007/s00211-015-0746-1](https://doi.org/10.1007/s00211-015-0746-1). Use for the velocity space, moments, unisolvence, and computable projection.
2. Beirão da Veiga, Mascotto, and Meng, *Interpolation and stability estimates for edge and face virtual elements of general order*, Mathematical Models and Methods in Applied Sciences 32 (2022), 1589–1631. [doi:10.1142/S0218202522500373](https://doi.org/10.1142/S0218202522500373). Use as supporting interpolation and stability literature; its space variants do not remove the need to state the estimates for the present interpolant.
3. Brenner, *Korn's inequalities for piecewise H^1 vector fields*, Mathematics of Computation 73 (2004), 1067–1087. [doi:10.1090/S0025-5718-03-01579-5](https://doi.org/10.1090/S0025-5718-03-01579-5). Use for the global broken Korn inequality.
4. Boffi, Brezzi, and Fortin, *Mixed Finite Element Methods and Applications*, Springer, 2013. [Publisher and DOI](#). Use for the continuous divergence lifting and mixed-method well-posedness.
5. Chen and Wang, *A divergence free weak virtual element method for the Stokes problem on polytopal meshes*, Journal of Scientific Computing 78 (2019), 864–886. [doi:10.1007/s10915-018-0796-5](https://doi.org/10.1007/s10915-018-0796-5). Use to identify the earlier weak VEM Stokes framework. A full comparison with stabilizer-free Stokes methods belongs in an introduction if one is added later.
6. Mu, Ye, and Zhang, *A stabilizer-free, pressure-robust, and superconvergence weak Galerkin finite element method for the Stokes equations on polytopal mesh*, SIAM Journal on Scientific Computing 43 (2021), A2614–A2637. [doi:10.1137/20M1380405](https://doi.org/10.1137/20M1380405). Add a short reference to this related stabilizer-free Stokes method.

Misprints and presentation

1. Define the symmetric strain, rigid motions, tensor trace, inner products, and mesh sizes before use. Distinguish a tensor's normal trace from its scalar matrix trace. Use K consistently in place of the unexplained E .
2. Use the existing tangential projection to shorten the weak-strain formula. Replace the two unconditional “ $\neq$ ” statements by an explanation that integration by parts also produces internal stress-jump terms.
3. Replace the hard-coded equation number 14 and the literal “problem (1)” with labels. Correct inconsistent boldface, missing integration measures, “degree of freedoms”, and repeated inequalities. Recall only the definitions needed by a proof; keep routine language plain and short.
4. Repair undefined commands in `numerics.tex`, use a proper numerical section heading, and give every table a label. The files `fig/poly`, `fig/tri`, and `fig/rect` are absent. Guard the figure input and state the missing-asset issue rather than inventing plots.

Author checks still needed

1. **Rectangular meshes are outside the stated proof.** For an axis-aligned rectangle, the face-tangential symmetric tensors span only the two diagonal matrices, not all of $\mathbb{S}$. Thus Assumption 2.1 fails. Keep the table as an empirical result, with this limitation stated. Extending the theory to rectangles would require additional analysis beyond this review. The geometry of the polygonal meshes must also satisfy the strengthened uniform assumption before the theorem is applied to them.
2. **Numerical reproducibility.** Supply the exact velocity, pressure, force, boundary values, viscosity, mesh construction, and definitions of the reported discrete L^2 and maximum norms. The source specifies only “sincosdata”. Confirm that the implementation uses the analyzed strain space: the original numerical section uses Σ_0^{-1} while Section 2 uses Σ_1 , and introduces 2ν whereas the model has coefficient one. The reported values are preserved, but their agreement with the corrected formulation cannot be certified from these files.
3. **Front matter.** The title command is commented out, the abstract and keywords are absent, and the submission date and funding statement are placeholders. These require author-supplied information; they are not filled in during proofreading.